GOPALAKRISHNAN THIRUNELLAI VENKITACHALAM

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EDUCATION

Carnegie Mellon University — August 2024 - May 2026
Master of Science (Research) — *CGPA: 4.0/4.0*
· Relevant Courses : Planning & Decision-Making for Robotics, Generative AI, Advanced NLP, 11-785: Deep Learning

Indian Institute of Technology Madras — October 2020 - June 2024
Bachelor of Technology (Honors) — Minor in Artificial Intelligence and Machine Learning — *CGPA: 8.71/10.00*
· Relevant Courses: Machine Learning, Reinforcement Learning, Robotics, Multi-Armed Bandits, Stochastic Processes

EXPERIENCE

Graduate Research Assistant (*advised by Prof. Maxim Likhachev*) — September 2024 - Present
Robotics Institute, CMU — *Pittsburgh, USA*

- **Formulated "Task-Metric Experience Graphs" (TMEG)**, a **C++** framework that adapts cached experience to dynamic constraints, solving the "static world" limitation of standard E-Graphs for contact-rich manipulation.
- **Engineered a Lipschitz-bounded heuristic** derived from the spectral norm of the Weighted Jacobian, guaranteeing mathematical consistency and admissibility for A^* search even when task targets shift.
- **Implemented high-performance search primitives** within the **SMPL** library, optimizing the sbpl codebase to support weighted $SE(3)$ metrics (Geodesic rotation) and enabling real-time warping of trajectories.
- **Deployed and validated the stack on a Ridgeback-UR10e manipulator** using **ROS/MoveIt**, achieving robust latch-disengagement in unstructured environments where standard planners failed.

PROJECTS

Algorithmic Stability & Convergence Analysis of Reinforcement Learning Systems

- **Implemented a dependency-free Reinforcement Learning library** in NumPy, implementing custom backpropagation kernels for **Dueling DQN** and **REINFORCE** to bypass high-level framework overhead and expose core gradient dynamics.
- **Validated manual gradient derivation pipelines** using SymPy, validating theoretical update rules against runtime behavior to ensure numerical precision in policy optimization steps.
- **Benchmarked algorithmic stability** in stochastic environments, isolating a **40% improvement in sample efficiency** for off-policy methods while quantitatively profiling variance sensitivity in noisy reward signals.

Generative Single-View 3D Reconstruction Pipeline [*Report*] [*Code*]

- **Engineered an end-to-end generative 3D pipeline** that fuses **Latent Diffusion inpainting** with **Monocular Depth Estimation (Depth-Anything-V2)** to autonomously synthesize complete, watertight 3D meshes from single RGB images.
- **Developed a novel iterative geometry repair algorithm** using **Open3D**, which dynamically detects low-density point cloud regions and applies Poisson reconstruction to resolve occlusions and eliminating geometric artifacts.
- **Achieved better performance vs TripoSR**, reducing Chamfer Distance by **55% (0.066 → 0.030)**and boosting structural consistency to **0.971**, validating high-quality asset generation.

Autonomous Manipulation & Planning Simulation Framework

- Designed a **high-performance Python middleware wrapping ManiSkill/SAPIEN**, abstracting complex physics engine interactions into a unified, deterministic **API** to **accelerate algorithm development and deployment**.[*code*]
- Built a **lightweight Multi-Agent Path Finding (MAPF)** engine from scratch, engineering custom collision kernels and data pipelines to reduce simulation overhead by removing external dependencies.[*code*]
- Developed an automated **benchmarking pipeline** to quantify planner performance, implementing rigorous regression testing for **theoretical optimality, cost convergence, and runtime efficiency across large-scale test suites**.

Constraint-Aware Path Planning for Large-Scale Robotics [*Code*]

- **Developed a highly optimized C++ formation control solver** utilizing the Eigen library for linear algebra operations, integrating Singular Value Decomposition (SVD) to enforce rigid-body constraints at **sub-millisecond latencies**.
- **Architected a real-time coordination algorithm** based on Priority Inheritance with Backtracking **(PIBT)**, enabling collision-free navigation for large-scale swarms in highly unstructured, obstacle-dense environments without gridlocking.
- **Solved the "formation fracture" problem** by designing a hierarchical cost function that dynamically weights geometric integrity against path efficiency, outperforming standard baselines in maintaining swarm cohesion under heavy congestion.

SKILLS

Languages	C++ (STL, Eigen), Python (PyTorch, NumPy, Pandas), MATLAB, Bash, JAX, Bazel
Robotics & Sim	ROS, MoveIt, Gazebo, Pybullet, Mujoco, IsaacSim, ManiSkill (SAPIEN), URDF/Xacro, RViz
Planning & Control	Graph Search (A*, ARA*), Sampling-based (RRT*, PRM), MAPF/PIBT, MPC
AI & Perception	RL (PPO, DQN), Stable Diffusion, Transformers, Open3D, OpenCV, Point Clouds
Developer Tools	Linux, Docker, Git, CMake